

Viscosity of deeply supercooled water and its coupling to molecular diffusion

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Edited by Pablo G. Debenedetti, Princeton University, Princeton, NJ, and approved August 12, 2015 (received for review May 7, 2015)

The viscosity of a liquid measures its resistance to flow, with consequences for hydraulic machinery, locomotion of microorganisms, and flow of blood in vessels and sap in trees. Viscosity increases dramatically upon cooling, until dynamical arrest when a glassy state is reached. Water is a notoriously poor glassformer, and the supercooled liquid crystallizes easily, making the measurement of its viscosity a challenging task. Here we report viscosity of water supercooled close to the limit of homogeneous crystallization. Our values contradict earlier data. A single power law reproduces the 50-fold variation of viscosity up to the boiling point. Our results allow us to test the Stokes–Einstein and Stokes–Einstein–Debye relations that link viscosity, a macroscopic property, to the molecular translational and rotational diffusion, respectively. In molecular glassformers or liquid metals, the violation of the Stokes–Einstein relation signals the onset of spatially heterogeneous dynamics and collective motions. Although the viscosity of water strongly decouples from translational motion, a scaling with rotational motion remains, similar to canonical glassformers.

supercooled water | viscosity | Stokes–Einstein relations

Water, considered as a potential glassformer, has been a long-lasting topic of intense activity. Its possible liquid–glass transition was reported 50 years ago to be in the vicinity of 140 K (1, 2). However, ice nucleation hinders the access to this transition from the liquid side. Bypassing crystallization requires hyperquenching the liquid at tremendous cooling rates, ca. $10^7 \text{ K} \cdot \text{s}^{-1}$ (3). As a consequence, many questions about supercooled and glassy water and its glass–liquid transition remain open (4–7).

As an example, crystallization of water is accompanied by one of the largest known relative changes in sound velocity, which has been attributed to the relaxation effects of the hydrogen bond network (8, 9). Indeed, whereas the sound velocity is around $1,400 \text{ m} \cdot \text{s}^{-1}$ in liquid water at 273 K, it reaches around $3,300 \text{ m} \cdot \text{s}^{-1}$ in ice at 273 K and a similar value in the known amorphous phases of ice at 80 K (10). Such a large jump is usually the signature of a strong glass, i.e., one in which relaxation times or viscosity follow an Arrhenius law upon cooling. However, pioneering measurements on bulk supercooled water by NMR (11) and quasi-elastic neutron scattering (12), as well as recent ones by optical Kerr effect (8, 9), reveal a large super-Arrhenius behavior between 340 and 240 K, similar to what is observed in fragile glassformers (13, 14). The temperature dependence of the relaxation time is well described by a power law (8, 9), as expected from mode-coupling theory (15, 16), which usually applies well to liquids with a small change of sound velocity upon vitrification. Based on these and other observations, it has been hypothesized that supercooled water experiences a fragile-to-strong transition (17). This idea has motivated experimental efforts to measure dynamic properties of supercooled water and has received some indirect support from experiments on nanoconfined water (18–20) and from simulations (21, 22).

In usual glassformers, many studies have focused on the coupling or decoupling between the following dynamic quantities: viscosity (η) and self or tracer diffusion coefficients for translation (D_t) and rotation (D_r). If objects as small as molecules were to follow macroscopic hydrodynamics, one would expect that the preceding quantities would be related through the Stokes–Einstein (SE),

$D_t \propto T/\eta$, and Stokes–Einstein–Debye (SED), $D_r \propto T/\eta$, relations, where T is the temperature. These relations are indeed obeyed by many liquids at sufficiently high temperature. However, they might break down at low temperature. Pioneering experiments were performed by the groups of Sillescu (23–25) and Ediger (26–28) where a series of molecular glassformers were investigated. SE relation is obeyed at sufficiently high temperature but violated around $1.3T_g$, where T_g is the glass transition temperature, thus indicating decoupling between translational diffusion and viscosity. In contrast, it was observed for ortho-terphenyl (23, 24, 26) that rotational diffusion and viscosity remain strongly coupled (i.e., obey the SED relation) even very close to T_g . A corollary is that translational and rotational diffusion decouple from each other at low temperature. These observations imply that deeply supercooled liquids exhibit spatially heterogeneous dynamics (29–31). Dynamic heterogeneities have been confirmed by direct observations of several single fluorescent molecules immersed in ortho-terphenyl (32) or nanorods immersed in glycerol (33). Physically different systems also show analogous behavior. Colloids near the colloidal glass transition violate SE but obey SED (34). In the metallic alloy $\text{Zr}_{64}\text{Ni}_{36}$, SE relation is even violated without supercooling, more than 35% above the liquidus temperature (35). This has also been related to the emergence of dynamic heterogeneities (36).

For water, SE already breaks down at ambient temperature, which corresponds to around $2.1 T_g$ ($T_g \simeq 136 \text{ K}$). Molecular dynamics simulations (37–39) have proposed that this occurs concurrently to dynamic heterogeneities caused by a putative liquid–liquid critical point. However, SE and SED also fail by application of high pressure at 400 K (40) where no liquid–liquid transition is expected. To gain more insight, the test of SE and SED in supercooled water deserves further investigation. Translational self-diffusion coefficient D_t (41) and rotational correlation time τ_r (assumed to scale as $1/D_r$) (42)

Significance

Water is the most ubiquitous liquid but also the most anomalous. In usual fluids far from their glass transition, viscosity and molecular diffusion are coupled through the Stokes–Einstein relations. For water, viscosity already starts decoupling from translational diffusion below 20°C. Simulations have suggested a connection with the putative separation of supercooled water into two distinct liquid phases. Whereas experimental diffusion data extend far in the supercooled region, accurate viscosity data were lacking due to the readiness of supercooled water to crystallize under the slightest perturbation. Using Brownian motion of spheres suspended in water, we have measured its viscosity down to -34°C without freezing. We find that whereas viscosity decouples increasingly from molecular translation upon cooling, it remains coupled to rotation.

Author contributions: F.C. designed research; A.D. performed research; B.I. and F.C. analyzed data; and B.I. and F.C. wrote the paper.

The authors declare no conflict of interest.

This article is a PNAS Direct Submission.

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This article contains supporting information online at www.pnas.org/lookup/suppl/doi:10.1073/pnas.1508996112/-DCSupplemental.

have thus been measured down to the homogeneous crystallization temperature (238 K) at ambient pressure. Their comparison reveals a decoupling between rotation and translation that increases with supercooling (42), similar to glassformers. However, viscosity data are needed for a direct test of SE and SED relations. Quite surprisingly, there are only two sets of data for the viscosity η at significant supercooling. Using Poiseuille flow in capillaries, Hallett (43) reached 249.35 K, and Osipov et al. (44) reached 238.15 K. However, the two sets disagree below 251 K, with an 8% difference at 249 K, beyond the reported uncertainties. The measurements in ref. 44 are suspected of errors (45) because of the small capillary diameter used. Here we report η at ambient pressure down to 239.27 K. Our study completes the knowledge of the main dynamic parameters of water down to the homogeneous crystallization limit and allows us to check the coupling of viscosity to molecular translation or rotation, as has been done for usual glassformers.

Results and Discussion

We studied the Brownian motion of polystyrene spheres with physical radius $a = 175 \pm 3$ nm using dynamic differential microscopy (46) (see *Materials and Methods* for details). In brief, a series of microscope images of a capillary containing the suspension of spheres in water is recorded. The image differences are analyzed to give the decorrelation time $\tau(q)$ of the light scattered by the suspension at a given wavenumber q . $\tau(q)$ decays as $1/q^2$, as expected for Brownian motion, and the coefficient gives the inverse of the translational diffusion coefficient D of the polystyrene spheres (Fig. 1, *Inset*). For these mesoscopic objects, the SE relation holds

$$D = \frac{k_B T}{6\pi\eta a_h}, \quad [1]$$

where k_B is the Boltzmann constant, T is the temperature, and a_h is the hydrodynamic radius of the sphere. Thus, our measurement of D allows us to deduce η . The uncertainty on η ranges from 2.3% at the highest temperature to 2.9% at the lowest temperature (*Materials and Methods*). We measured four different samples which were able to reach 239.43, 244.45, 241.86, and 239.27 K without crystallization. The results are shown in Fig. 1, Fig. S1, and Tables

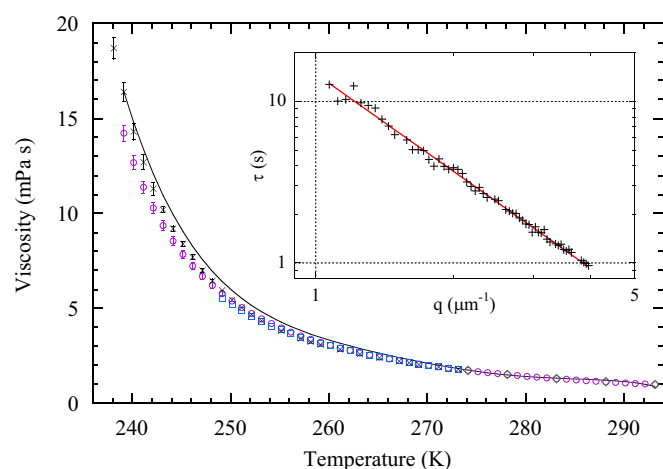


Fig. 1. Viscosity of water: data from this work (purple circles), Hallett (43) (blue squares), Collings and Bajenov (70) (green diamonds), and Osipov et al. (44) (black crosses). The error bars are only displayed below 249 K for clarity. The curve shows the calculated effective viscosity (based on our data) that would be measured by Poiseuille flow in a capillary with radius $1 \mu\text{m}$ (44) and surface potential $\phi = 4$ mV (see *Materials and Methods* for details). (*Inset*) Image decorrelation time $\tau(q)$ (pluses) showing a $1/(Dq^2)$ dependence (red line). The data shown are for run 1 at 239.43 K.

S1 and S2 and compared with literature values in Fig. S2. Good agreement is found for the stable liquid above 273 K. We confirm Hallett's measurements on supercooled water (43) and extend them by 10 K. We find however systematically lower values than Osipov et al. (44) at low temperature.

Referring to previous work (47) showing an increase of the density of water inside small μm -sized capillaries, Cho et al. (45) argued that this effect might have changed the viscosity of water in the experiment of Osipov et al., who used quartz capillaries with radius $R = 1 \mu\text{m}$. We propose another possible bias: electroosmosis. The quartz surface being charged, it induces a layer of counter ions in the liquid close to the surface. Imposing mass flow through the capillary generates a current, which in turn creates a voltage opposing the flow (48). This leads to an increased viscosity compared with the bulk one. Detailed calculations are given in *Materials and Methods*. The main ingredient is the surface potential ϕ on the capillary walls. We found that a temperature-independent value $\phi = 4$ mV is sufficient to increase our viscosity values to those of ref. 44. The effect scales as $1/R^2$, which explains why it did not affect the data of Hallett, who used $R = 100 \mu\text{m}$.

To analyze the data over a broad temperature range, we have combined our data with literature values, selecting in each temperature interval among the most accurate data available (see *Supporting Information* and Tables S3 and S4 for details). Fig. 2 compares different usual parameterizations of the temperature variation of viscosity from the boiling point to the maximum supercooling temperature. We start with two non-diverging functional forms of the viscosity. The first is the venerable Arrhenius law [$\eta(T) = \eta_0 \exp[E_a/(k_B T)]$], obeyed by strong glassformers. It reflects thermally activated transport over the energy barrier E_a . For water (Fig. 2), $\ln \eta$ varies more rapidly than inverse temperature, indicative of a fragile behavior. The second is the parabolic law [Arrhenius law above T_0 switching to $\eta(T) = \eta_0 \exp[J^2(1/T - 1/T_0)^2 + E_a/(k_B T)]$ below T_0] introduced by Elmatad et al. (49). They were able to collapse with this law transport properties of 58 fragile glassformers and also simulation data, including several models of water (50). However, when applied to real water (see *Supporting Information* for details about the fitting procedure), the parabolic law is not able to reproduce the data correctly, showing deviations much larger than the experimental uncertainties at high and low temperatures. In addition, in contrast to the analysis with the parabolic law of simulated structural relaxation time for five water models (50), $T_0 = 305.15$ K is significantly above the experimental temperature of maximum density (277.14 K), and the parameter J/T_0 is 3.6 instead of 7.4 ± 0.4 for simulations.

Next we test two models with an apparently diverging viscosity. The first is the Vogel–Fulcher–Tammann (VFT) equation, $\eta(T) = \eta_0 \exp[BT_0/(T - T_0)]$. This originally empirical fit can be interpreted within the Adam–Gibbs theory (51), which identifies T_0 as the Kauzmann temperature. At T_0 , the supercooled liquid would have the same entropy as the crystalline phase. T_0 would however never be reached because it lies below T_g . The VFT fit performs slightly better than the parabolic law but falls beyond the experimental uncertainties at high and low temperatures. In addition, the fit value for $T_0 = 168.9$ K is above $T_g \approx 136$ K for water, which is not in line with the Adam–Gibbs picture. Finally, we consider a power law, $\eta(T) = \eta_0 (T/T_s - 1)^{-\tau}$. This form is suggested by two physical pictures. One is mode-coupling theory (15, 16), with T_s identified as the mode-coupling temperature $T_s = T_c > T_g$. Obviously, viscosity cannot diverge above T_g ; this divergence would be avoided thanks to the thermally activated process of hopping of molecules, which would lead to a departure from the power law slightly above T_s (16). The other picture was proposed by Speedy and Angell (52), who noticed that many thermodynamic and dynamic properties of water seem to have a singular behavior, extrapolating via power laws to a divergence at a common temperature T_s . They proposed two

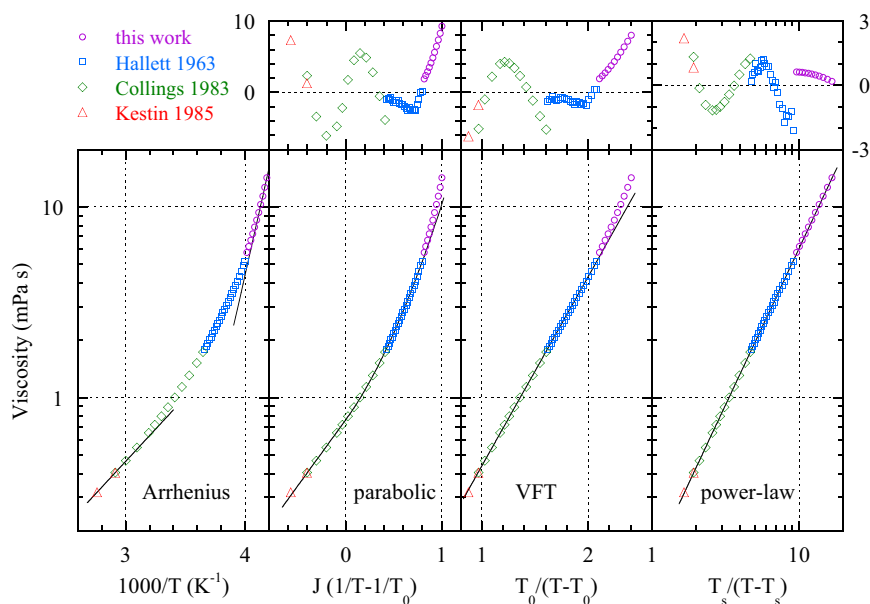


Fig. 2. Parameterizations of viscosity. The four panels include four datasets: this work (purple circles) and refs. 43 (blue squares), 70, (green diamonds), and 71 (red triangles). (Left) Arrhenius plot, showing an apparent activation energy increasing from 1,560 to 6,410 K upon cooling (solid lines). (Left Center) Parabolic law, with best fit ($\chi^2 = 14.2$) parameters $T_0 = 305.15$ K, $\eta_0 = 2.323 \cdot 10^{-6}$ Pa s, $J = 1,112$ K, $E_a = 1,769$ K. (Center Right) VFT representation, with best fit ($\chi^2 = 10.5$) parameters $T_0 = 168.9$ K, $\eta_0 = 4.442 \cdot 10^{-5}$ Pa s and $B = 2.288$. (Right) Power law representation, with best fit ($\chi^2 = 0.91$) parameters $T_s = 225.66 \pm 0.18$ K, $\eta_0 = (1.3788 \pm 0.0026) \cdot 10^{-4}$ Pa s, and $\gamma = 1.6438 \pm 0.0052$. (Top) The normalized residuals $(\eta_{\text{exp}} - \eta_{\text{fit}}) / \sigma_{\text{exp}}$, where η_{exp} and η_{fit} are the experimental and fitted viscosity, respectively, and σ_{exp} is the experimental uncertainty (1 SD). Note that the vertical scale of Top Right is different.

interpretations: the existence of “a λ transition associated with the cooperative formation of an open hydrogen-bond network” or a “limit of mechanical stability for the supercooled liquid phase.” The power law gives an excellent fit over the whole 134-K interval. It would extrapolate to a singular temperature $T_s = 225.66 \pm 0.18$ K. This is above T_g but slightly below the homogeneous crystallization temperature T_h , so that a true divergence cannot be directly observed by experiment. The quality of the power law fit was noted earlier but using data down to 249 K only (52). A power law also gives the best fit to D_t up to 500 K and to τ_r up to 450 K but with lower values of T_s (see Figs. S3 and S4, Table S5, and Supporting Information for details about the error bars on the fit parameters). This is at odds with mode-coupling prediction of a common T_s values for all dynamic properties or with the assumption of a common singularity temperature.

We now turn to the discussion of the coupling between viscosity and molecular translation and rotation. In Fig. 3, we test the SE and SED relations by plotting as a function of T the quantities $D_t\eta/T$ and $\eta/(\tau_r T)$, which should be constant if SE and SED relations hold. This is the case at high temperature, but the relations start being violated below around 350 K. The violation of SE is stronger, reaching 70% at 239 K, with an increasing trend as the temperature decreases. The maximum violation of SED is only around 18% at 239 K. This behavior is reminiscent of what is observed for ortho-terphenyl (23, 24, 26), although the viscosity range investigated for water is more modest. We also emphasize once more that SE violation in water occurs far above T_g . Our findings contrast with molecular dynamic simulations (MDS) of the ST2 model for water (37, 38), which find that SED is violated even more strongly than SE in the same temperature range. The reason for the discrepancy may lie in the different quantities used to check SED. Indeed, NMR experiments access τ_r only, whereas MDS can access both τ_r and the rotational diffusion coefficient D_r of the water molecules and find that the usual assumption $D_r \propto 1/\tau_r$ is not valid (37, 38). Using D_r , ST2 MDS find that SED holds at high temperature and is violated upon cooling. If $1/\tau_r$ is used instead, SED is never valid. However, we note that MDS also use a proxy for the viscosity: they replace it by the alpha-relaxation time τ_α , assuming $\eta \propto \tau_\alpha$. In a careful study of model atomic and molecular systems (53), it was found that τ_α/η is temperature-dependent, which causes biases in the evaluation of SE and SED using τ_α . Therefore, it appears highly desirable to determine both η and τ_r in MDS of a water model, to compare with the available experimental data.

Several models have attempted to improve on SE. A simple hydrodynamic model (54) treating water molecules as spheres with radius a moving into a surrounding continuous medium introduces a slip boundary condition, which modifies the proportionality coefficients between D_t , D_r , and T/η . This model describes well the data above 360 K, with a molecular diameter $2a = 0.25$ nm and a slip length of $0.23a$. However, the low-temperature data analyzed with this model yield an unphysical 60% decrease of a . Another explanation of the departure from SE and SED is given by mode coupling theory (16), which predicts a universal behavior: $\eta \propto 1/D_t \propto \tau_r$. Although this should only occur close to the mode coupling temperature T_c (typically $1.3T_g$), experiments on $\text{Zr}_{64}\text{Ni}_{36}$ (35) have found a constant $D_t\eta$ up to $2T_c$, 500 K above the liquidus temperature (1283 K). $D_t\eta$ is also found to be nearly constant for water between 259 and 300 K. However, when our data are included, $D_t\eta$ increases sharply between 259 and 249 K, whereas η/τ_r levels off (Fig. S5). The observed behavior of $D_t\eta$ thus disagrees with the mode coupling predictions.

Other modifications of SE and SED are the fractional SE (FSE) and fractional SED (FSED) relations, with D_t and τ_r proportional to $(\eta/T)^\zeta$. Such relations or close variants were originally introduced for tracer diffusion in various stable liquids (55, 56) and for diffusion in supercooled ortho-terphenyl (23, 24). In the latter case, the FSE exponent exhibits a crossover from $\zeta = -1$ at high temperature to $\zeta = -0.79$ at low temperature. A similar crossover of the FSE exponent was observed for Tris-naphthylbenzene from $\zeta = -1$ to $\zeta = -0.77$ (27). The FSE relation has since been tested for a variety of liquids and found to give a good description far from the glass transition (57). A putative universal crossover from FSE with $\zeta = -1$ to $\zeta \simeq -0.85$ upon cooling was suggested by the analysis of data for nine liquids (58). Simulations of models of supercooled fragile liquids (East models) find $\zeta \simeq -0.7$ to -0.8 (59, 60).

Fig. 4 investigates the validity of FSE and FSED relations for water. In the case of translation, a transition between two FSE relations with exponents 0.94 and 0.67 has been proposed earlier (57, 61) but was based on the flawed data from ref. 44. Using our new data, we see that a single exponent ζ is still not sufficient. The limited experimental range does not allow deducing reliable exponents at high and low temperatures. However, the data are consistent with the ST2 MDS (37, 38), which find a crossover between $\zeta = -1$ at high temperature and $\zeta = -0.8$ at low temperature. MDS of another water model (mW) find a crossover from $\zeta = -1$ to $\zeta = -0.75$ (62). The overall picture is that a similar crossover in

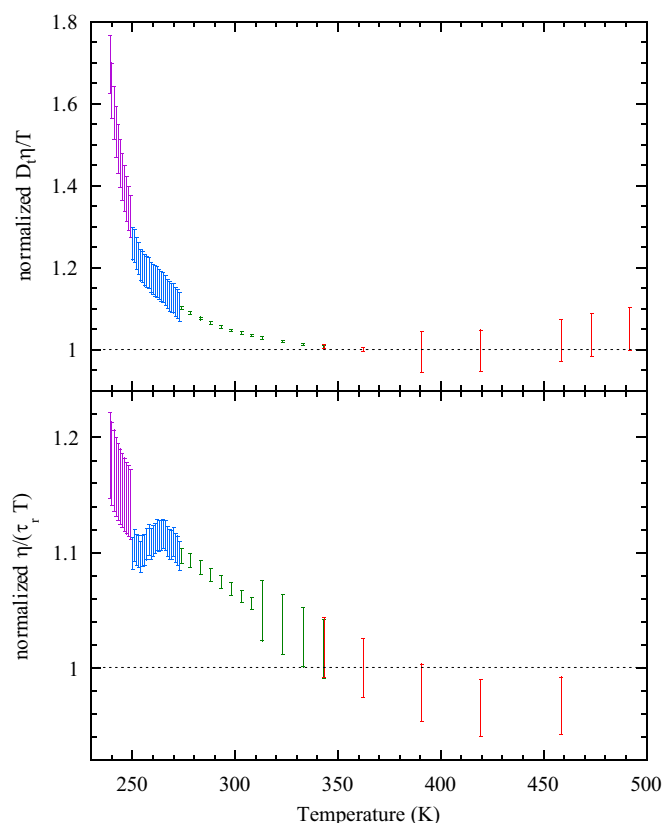


Fig. 3. Test of the Stokes–Einstein and Stokes–Einstein–Debye relations. $D_t \eta / T$ (Upper) and $\eta / (\tau_r T)$ (Lower) are plotted as a function of temperature. The viscosity data used are the same as in Fig. 2 (same colors), and D_t and τ_r were calculated at the temperatures of the viscosity data using the power law fits given in Table S5. Only the combined uncertainty (1 SD) without the data symbol is displayed for clarity. The data were further normalized by their value at 362.25 K. SE and SED relations would thus correspond to the horizontal dotted lines. SE and SED hold at high temperature, but they are violated by around 70% and 18% at low temperature, respectively.

the FSE exponent is observed for real water, simulations of water models (37, 38, 62), and experiments on ortho-terphenyl (23, 24), Tris-naphthylbenzene (27), and seven other liquids (58).

In the case of rotation, we find that FSED experimentally holds with a single exponent $\zeta = 0.97$ over more than 2 decades. We emphasize that this would not be the case if the low-temperature data from Osipov et al. (44) were used instead of ours (Fig. 4). The success of FSED for experiments contrasts with the ST2 MDS results, which find a crossover between two FSED relations with different exponents. However, as noted above, more simulations, calculating η rather than τ_α , are needed to allow a direct comparison. As for FSE, the experimental result for FSED in water resembles that for ortho-terphenyl where $\zeta = 1$ (23, 24). We also note that a crossover from FSED with $\zeta = 1$ to $\zeta \approx 0.85$ upon cooling was suggested by the analysis of data for six liquids (58); however, the structural relaxation time was used instead of τ_r , preventing a direct comparison.

When the coupling between dynamic quantities is considered, water behaves similarly to canonical glassformers. The close analogy with ortho-terphenyl is striking. However, a major difference in the case of water is that SE violation already occurs far above T_g . By analogy with what is known for molecular glassformers, this suggests the existence of dynamic heterogeneities in water. They have been observed in MDS (37–39) which were discussed in the context of a liquid–liquid transition, which exists in at least one of the interaction potential studied, ST2 (63). However, we note that

MDS of another water-like model (mW) for which there is no liquid–liquid transition (64) find a violation of SE relation and crossover to a FSE relation with $\zeta = -0.75$ (62), in the temperature range where a growing correlation length is detected (65). Various water models thus predict correlated regions whose size increases upon cooling. Experiments have found that SE and SED in water are also violated at high temperature by application of high pressure (40). In that case it was attributed to the rigidity of the first neighbors shell in water and to the invariant number of hydrogen bonds at high pressure. What would be the effect of pressure on SE and SED at the lowest temperatures? MDS of a water model with a liquid–liquid transition have investigated SE breakdown in the supercooled liquid under pressure (39). They find that the temperature at which SE starts being violated correlates with the locus of heat capacity maxima emanating from the liquid–liquid critical point seen in the simulations. Experiments are underway to extend our viscosity measurements to higher pressure.

Materials and Methods

Samples. We use a dispersion of polystyrene spheres (Duke Scientific; mean physical diameter 350 ± 6 nm) in water, placed in a rectangular borosilicate capillary (Vitrotubes; 20 μm thick and 200 μm wide). The capillary is glued with epoxy glue (Araldite; Huntsman Advanced Materials) at both ends, to prevent any advective motion of the spheres (e.g., induced by Marangoni effects). The spheres are small enough to exhibit Brownian motion, and their density close to 1 prevents settling on the experimental timescale. The volume fraction of the

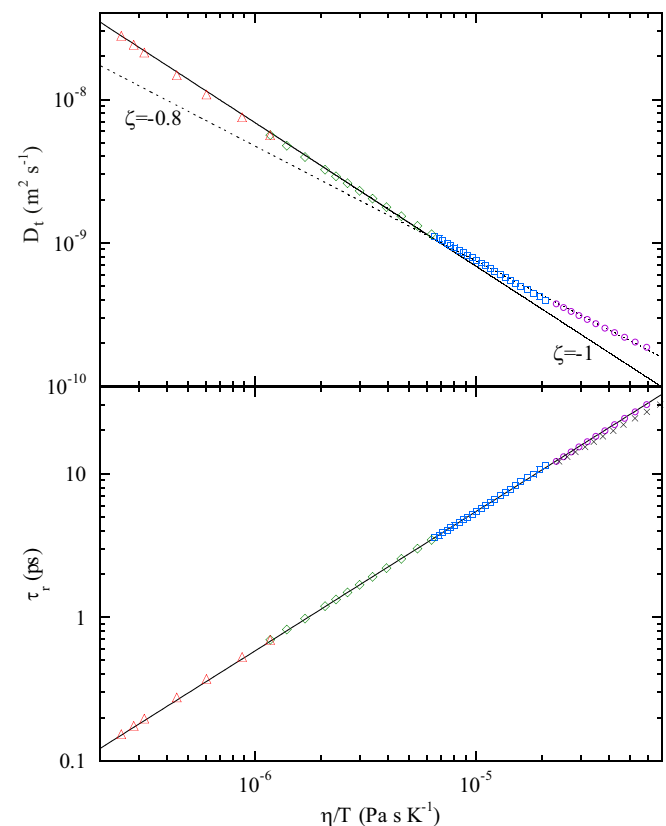


Fig. 4. Test of the fractional Stokes–Einstein and Stokes–Einstein–Debye relations. D_t (Upper) and τ_r (Lower) are plotted as a function of η / T . The viscosity data used are the same as in Fig. 2 (same symbols and colors), and D_t and τ_r were calculated at the temperatures of the viscosity data using the power law fits given in Table S5. FSE (Upper) does not hold with a single exponent ζ , but the data suggest a crossover from $\zeta = -1$ to -0.8 (solid lines) upon cooling, as observed in ST2 MDS (37). FSED relation (Lower) holds with $\zeta = 0.97$ (solid line). The low temperature data from Osipov et al. (44) (black crosses) deviate from the FSED fit.

commercial dispersion is 1%. We checked that the dilution of the particles had no influence on the measurements. However, the more diluted they were, the larger supercooling that could be reached. Thus, for a good balance between signal and supercooling, we used 100 times dilutions (0.01% volume fraction), which allowed reaching around 239 K in two capillaries. Below that temperature the sample systematically crystallized before the measurements could be performed. The effect of the purity of the solution in which the spheres are suspended was also checked, by centrifugating a sample and replacing the supernatant with ultrapure water (Direct-Q3 UV; Millipore) several times. This did not change the results.

Experimental Setup. The capillary filled with the suspension of spheres is glued on a microscope slide placed on the Peltier temperature stage (Linkam LTS 120) of an upright microscope (Zeiss AxioScope) equipped with a $\times 100$ long-working distance objective (Mitutoyo Plan Apo, NA 0.7). The numerical aperture of the light source was chosen as small as possible ($N_A = 0.1$), and the diaphragm aperture was maximum. The intensity of illumination had no influence on the results. To calibrate the actual temperature inside the capillary down to 239 K, we used the same capillaries as for the viscosity measurement but filled with pure chemicals whose known melting points (66) served as reference. The temperature calibration was repeated just after each of the runs reported in this work. The temperature uncertainty is 0.15 K. The spheres' motion was recorded with a CCD camera (Prosilica GX1050, $1,024 \times 1,024$ pixels²; Allied Vision Technologies) able to reach up to 112 fps. The frequency of the camera was checked by filming a LED flashing with a tunable frequency. No significant deviation was observed between the imposed flashing frequency and the one deduced from the movies (less than 0.01% at 100 fps).

Dynamic Differential Microscopy. Typical data consist of a sequence of 500 images, acquired at 10–100 fps depending on temperature, with 8 ms exposure time. Images are processed using a home-written code (MATLAB; Mathworks) based on Fourier analysis as explained in ref. 46. This yields the decorrelation time τ as a function of the wave vector q . The diffusion coefficient D for a given movie was obtained by least-squares fitting $y = \ln \tau$ as a function of $x = \ln q$ with the function $y = -\ln D - 2x$. A typical curve is shown in Fig. 1, *Inset*. All movies were analyzed with the same q interval. Changing the boundaries by 10% does not affect the results.

Viscosity Values. To convert the values of D measured by dynamic differential microscopy into the viscosity η of water, the starting point is the Stokes–Einstein relation for a Brownian sphere (Eq. 1). We have converted our D values into viscosity η using

$$\eta(T) = \eta(T_0) \frac{T}{T_0} \frac{D(T_0)}{D(T)}, \quad [2]$$

where $T_0 = 293.15$ K is a reference temperature at which the viscosity is known with high accuracy (67): $\eta(T_0) = 1.0016 \pm 0.0017$ mPa s. The experimental value for $D(T_0)$ was set by repeating the measurement at T_0 in 12 independent capillaries, which gave $D_0 = 1.086 \pm 0.027 \mu\text{m}^2 \cdot \text{s}^{-1}$. The final viscosity measurements were performed in four independent runs in four different capillaries, with 1–11 movies per capillary (typically 3–5) at each temperature. The detailed viscosity dataset and smoothed values are given in Tables S1 and S2, respectively.

Hydrodynamic Radius. As a further check of the procedure, we can also calculate with Eq. 1 the hydrodynamic radius a_h . At T_0 we find $a_h = 189.1 \pm 4.0$ nm based on the 12 independent measurements. This is slightly higher than the value given by the manufacturer: 179 ± 5 nm. This small difference can be explained by the effect of confinement on Brownian motion. Several corrections have been proposed for a sphere of radius a_h confined between two parallel walls separated by a distance d (68). All corrections are similar for the large value of $d/(2a_h) \simeq 50$ (weak confinement) corresponding to our experiment. To estimate the effect, we use the Oseen formula (equation 1a in ref. 68). For a sphere whose center is at a distance z from one of the confining walls,

$$\frac{D_{\parallel}}{D_{SE}} = 1 - \frac{9a_h}{16} \left(\frac{1}{z} + \frac{1}{d-z} \right), \quad [3]$$

where $D_{\parallel}$ is the parallel diffusion coefficient in confinement and D_{SE} is the bulk diffusion coefficient given by Eq. 1. The experiment averages between $z = a_h$ and $z = d - a_h$, which yields

$$\frac{D_{\parallel}}{D_{SE}} = 1 - \frac{9}{8} \frac{a_h}{d - 2a_h} \ln \left(\frac{a_h}{d - a_h} \right). \quad [4]$$

This relation combined with the experimental values of D_0 yields $a_h = 179.9 \pm 3.7$ nm in excellent agreement with the manufacturer value. This validates the procedure. It is important to note that the correction factor to D_{SE} in Eq. 4 is independent of viscosity and therefore of temperature. The thermal expansion of polystyrene is also negligible in the temperature range investigated. This justifies the use of Eq. 2 to convert D into η at all temperatures. The validity of our approach is further corroborated by the excellent agreement with Hallett's data which cover the range 239.15–273.15 K (Fig. 1 and Fig. S2).

Measurement Uncertainty. We take for the intrinsic relative uncertainty (1 SD) on η the 2.3% SD of the measurements on 12 independent capillaries at $T_0 = 293.15$ K. Because the viscosity is well described by a power law $[\eta(T) = \eta_0(T/T_s - 1)^{-\gamma}]$, the effect of the 0.15 K temperature uncertainty was taken into account to give the total relative uncertainty (1 SD) at temperature T using $\{(0.023)^2 + [0.15\gamma/(T - T_s)]^2\}^{1/2}$. The resulting uncertainty ranges from 2.3% at the highest temperature to 2.9% at the lowest temperature. Analysis of the data scatter around the smoothed values (Fig. S1) is fully consistent with this calculated uncertainty.

Electroosmotic Effect. To explain the discrepancy of Hallett's and our data with the data of Osipov et al. (44), we propose that the latter measurements were affected by an electroosmotic effect. Viscosity was deduced from a Poiseuille flow in a 1- μm -radius quartz capillary. The surface charges present at the quartz surface are advected by water, creating an electric current in the capillary. As this capillary is an open electric circuit, a voltage appears between the ends of the capillary that cancels this electric current. The resulting fluid flow is thus a combination of the pressure gradient and the electric field (48). It can be shown that this leads to the measurement of an effective viscosity

$$\eta_{\text{eff}}(T) = \frac{\eta(T)}{1 - \frac{a(T)}{\eta(T)}}, \quad [5]$$

with

$$a(T) = \frac{8\epsilon_0^2}{R^2} \frac{\epsilon^2(T)\phi^2(T)}{\sigma(T)} > 0, \quad [6]$$

where R is the radius of the capillary, ϵ is the relative dielectric permittivity of water, ϕ is the electric potential of the capillary surface, and σ is the electrical conductivity of the fluid. The viscosity deduced with usual Poiseuille flow formulas is thus overestimated. The effect decreases with the radius of the capillary as $1/R^2$ and becomes negligible for wide capillaries, such as used by Hallett ($R = 100 \mu\text{m}$).

To model the electroosmotic effect in supercooled water, we extrapolate formulas for the conductivity (69) $\sigma(T)$ and the permittivity (66) $\epsilon(T)$ of stable water:

$$\sigma(T) = \sigma_0 \exp \left(-A - \frac{B}{T} - \frac{C}{T^2} - \frac{D}{T^3} \right), \quad [7]$$

with $\sigma_0 = 100 \mu\text{Sm}^{-1}$, $A = -1.7756$, $B = -903.75$ K, $C = 0.4134010^6$ K², and $D = 0.8104610^8$ K³, and

$$\epsilon(T) = \epsilon_1 + \epsilon_2 T + \epsilon_3 T^2, \quad [8]$$

with $\epsilon_1 = 249.21$, $\epsilon_2 = -0.79069$ K⁻¹, and $\epsilon_3 = 7.299710^{-4}$ K⁻².

Finally, we combine Eqs. 5–8 and compute the effective viscosity from our measured viscosities, looking for the value of the surface potential ϕ that reproduces Osipov's data. Assuming a temperature-independent ϕ for simplicity, we find $\phi = 4$ mV (Fig. 1), a reasonable value.

ACKNOWLEDGMENTS. We thank Marine Borocco for her help at the early stage of the experiments, Luca Cipelletti and Cécile Cottin-Bizonne for discussion about dynamic differential microscopy, Abraham D. Stroock and Anne-Laure Biance for discussion about electrokinetics, and Livia E. Bove for a critical reading of the manuscript. We acknowledge funding by the European Research Council under the European Community's FP7 Grant Agreement 240113 and by the Agence Nationale de la Recherche Grant 09-BLAN-0404-01.

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